Fermi motion in untagged J/ψ photoproduction on the deuteron

Bremsstrahlung photon beam from the 5 cm target, spectator model of the deuteron, and the yield ratio to a free neutron at rest

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generator: `exclurad_py`, branch `jpsi-fermi`, module `generator/photo.py`

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Abstract

A 10.6 GeV electron beam radiates in the 5 cm liquid-deuterium cell of CLAS12 and the bremsstrahlung photons produce J/ψ mesons on the neutrons of the deuteron, $\gamma n \to J/\psi\, n$. We describe the photon spectrum used, including the correct behaviour at the endpoint $E_\gamma \to E_0$ where the complete-screening formula fails, the spectator model of the deuteron with the AV18 momentum distribution, and the Monte Carlo that combines them with the J/ψ cross section of the `roberts2026` parametrisation. With 20 million events per sample the ratio of the J/ψ yield per neutron on the deuteron to the yield on a free neutron at rest is measured to 0.2 % per 100 MeV bin of photon energy. Above $E_\gamma = 8.6$ GeV the ratio stays between 0.9889 and 1.0173, with a mean of 0.9944 ± 0.0003: Fermi motion changes the yield per neutron by less than 1 %. Two effects of comparable size and opposite sign make up this number: the binding of the struck neutron lowers the available γn energy and reduces the yield by about 4 %, and the Møller flux factor of the moving neutron, correlated with the steeply rising cross section, raises it by about the same amount. Fermi motion is therefore negligible for the yield above 8.6 GeV, but the shape of the ratio near the free threshold at 8.20 GeV and the yield below it, 2.55 % of the total, are entirely due to it. It also dominates the resolution of every quantity reconstructed with the target assumed at rest: the photon energy obtained from the J/ψ alone is smeared by 196 MeV rms and shifted by up to +120 MeV near threshold. The conclusion (Section 8.1) is that the Monte Carlo has to include Fermi motion for any analysis on deuterium, although the generated yield per neutron above 8.6 GeV does not require it. Section 7 shows that the cross section must be presented against W from the final state, with the lab photon energy, reconstructed exactly from the spectator's mass-shell condition, kept as a label: one photon energy feeds a 0.13 GeV band in W, sub-threshold photons produce the reaction at an ordinary W of 4.2 GeV, and the effective photon flux per W bin, which the generator provides, differs from the free one by 2–3 % below 9 GeV and by up to 40 % at the endpoint.

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1 Purpose and set-up

The question asked was simple: how much does the Fermi motion of the neutron in the deuteron change the J/ψ photoproduction yield in the CLAS12 configuration, where the photon beam is the untagged bremsstrahlung of the 10.6 GeV electron beam in the target itself? The scattered electron is not observed; it stays in the beam line. The observable is the $J/\psi \to e^+e^-$ yield as a function of the photon energy reconstructed from the final state, and its t distribution.

The generator therefore has three ingredients, described in the next three sections:

1. the photon spectrum dN_γ/dE_γ per incident electron, from thin-target bremsstrahlung in the cell, with the endpoint treated correctly (Section 2);

2. the deuteron, in the plane-wave spectator model with the AV18 momentum distribution (Section 3);

3. the elementary cross section $\sigma(\gamma N \to J/\psi N)$ and its t dependence (Section 4).

Section 5 gives the algorithm and its validation, Section 6 the results of the high-statistics samples, and Section 8 the interpretation and the limits of the model. All samples were produced with `exclurad_py` on the `jpsi-fermi` branch; the cards and commands are in Appendix A.

Three samples are compared throughout, all with the same photon spectrum and the same elementary cross section:

sample	target	events
deuteron	neutron of the deuteron, AV18 Fermi motion, spectator proton written out	20000000
neutron	free neutron at rest	20000000
proton	free proton at rest (LH_2 radiator thickness), for cross-checks	20000000

The *deuteron/neutron* ratio, yield per neutron, isolates the effect of Fermi motion and binding; the photon flux and the cross section cancel in it bin by bin.

2 The photon beam: bremsstrahlung in the target

2.1 Thin-target spectrum

The beam electron of energy $E_0 = 10.6\,\mathrm{GeV}$ traverses the liquid target and radiates. The 5 cm cell is $L/X_0 = 5/769.1 = 6.5 \times 10^{-3}$ radiation lengths of LD_2 (5.6×10^{-3} for LH_2, $X_0 = 890.4\,\mathrm{cm}$), so the thin-target formula applies: the photon spectrum is linear in the thickness and the energy loss of the electron before it radiates is neglected (it is at most 1.7 MeV over the full cell, see Section 2.2). We use the complete-screening form of Tsai,

$$\frac{dN_\gamma}{dE_\gamma} = \frac{X}{X_0}\,\frac{1}{E_\gamma}\Big[\tfrac{4}{3}\,(1-y) + y^2\Big]\,S(E_\gamma), \qquad y = \frac{E_\gamma}{E_0}, \tag{1}$$

where X is the thickness of material the electron has crossed before the production vertex and $S(E_\gamma)$ is the endpoint factor of Section 2.2, equal to one except in the last $\sim$300 MeV. The definition of X_0 absorbs the atomic screening logarithms and the contribution of the atomic electron, which for hydrogen and deuterium is as large as that of the nucleus.

The photon flux grows linearly along the cell, so the production vertices are distributed with a density proportional to the upstream thickness, $dP/dz \propto z - z_{\mathrm{in}}$; the generator draws $z = z_{\mathrm{in}} + L\sqrt{u}$ with u uniform. Integrated over the cell, the yield per incident electron and per target nucleon is that of the average thickness, $X = L/2$, and this is the factor used in the event weight,

$$\frac{X}{X_0} = \frac{L}{2X_0} = 3.25 \times 10^{-3}\ (\mathrm{LD_2}), \qquad 2.81 \times 10^{-3}\ (\mathrm{LH_2}). \tag{2}$$

With these numbers the LD_2 cell delivers 1.439×10^{-3} photons per electron between 6.5 and 10.6 GeV, and 0.768×10^{-3} above the free-neutron threshold of 8.20 GeV (Figure 1, left).

2.2 The endpoint

Formula (1) without S is finite at $E_\gamma = E_0$: the bracket tends to one and the spectrum to $1/E_0$. This is an artefact of the complete-screening approximation, which assumes that the minimum momentum transfer to the atom,

$$q_{\min} = \frac{m_e^2\,E_\gamma}{2E\,E'}, \qquad E = E_0,\ E' = E_0 - E_\gamma, \tag{3}$$

is small compared with the inverse screening radius $1/a_s$ of the atomic electron cloud. As $E' \to 0$ the momentum transfer grows without bound, screening switches off, and the cross section is governed by the unscreened logarithm $\ln(2EE'/m_eE_\gamma)$, which falls through zero when E' becomes of the order of m_e. **The spectrum therefore does go to zero at the endpoint**, over the last fraction of an MeV, after bending down over the last few tens of MeV.

We use the analytic intermediate-screening spectrum of Schiff (Koch and Motz formula 3BS(b)), which interpolates between complete and no screening with the momentum transfer of each photon energy. With $r = E'/E$ and, in units of m_e,

$$b = \frac{2EE'}{a_s\,E_\gamma}, \qquad \frac{1}{M(0)} = \Big(\frac{E_\gamma}{2EE'}\Big)^2 + \frac{1}{a_s^2}, \tag{4}$$

Schiff's spectrum is

$$\begin{aligned}\frac{d\sigma}{dE_\gamma} \propto \frac{1}{E_\gamma}\Big\{&\Big[1 + r^2 - \tfrac{2}{3}r\Big]\Big[\ln M(0) + 1 - \tfrac{2}{b}\arctan b\Big] \\ &+ r\Big[\tfrac{2}{b^2}\ln(1+b^2) + \tfrac{4(2-b^2)}{3b^3}\arctan b - \tfrac{8}{3b^2} + \tfrac{2}{9}\Big]\Big\}.\end{aligned} \tag{5}$$

Its complete-screening limit $b \to \infty$ is $\frac{1}{E_\gamma}\{[1+r^2-\frac{2}{3}r](2\ln a_s+1)+\frac{2}{9}r\}$, and $1+r^2-\frac{2}{3}r = \frac{4}{3}(1-y)+y^2$ is exactly the Tsai bracket. The endpoint factor is the ratio

$$S(E_\gamma) = \frac{\text{Eq. (5)}}{\text{its complete-screening limit}}, \tag{6}$$

cut at zero where Schiff's bracket turns negative, and it multiplies the Tsai shape in Eq. (1). Applying the ratio rather than Eq. (5) itself keeps the normalisation of X_0, which contains the electron contribution and the Coulomb-free $Z = 1$ logarithms, and changes only the shape where the physics changes.

The screening radius is the one parameter. Schiff's Thomas–Fermi value $a_s = 111\,Z^{-1/3}$ (in Compton wavelengths) describes many-electron atoms; hydrogen and deuterium have a single $1s$ electron and the physical screening length is the Bohr radius, $a_0 = \lambda_C/\alpha = 137$ Compton wavelengths. We use 137; Figure 1 (right) shows that the difference from 111 is confined to the last few MeV. The resulting factor is

$E_0 - E_\gamma$ [MeV]	300	100	30	10	3	1
$S(E_\gamma)$	0.968	0.910	0.769	0.582	0.362	0.160

and $S = 0$ for $E_0 - E_\gamma < 0.42\,\text{MeV}$. The endpoint correction removes 1.24 % of the photons between 6.5 and 10.6 GeV. Its effect on the J/ψ yield is proportionally larger than on the photon count, because the cross section is largest at the highest photon energies, but it is still a sub-percent effect on the total; what it does change is the last 100 MeV of the E_γ spectrum, which is what a measurement near the endpoint sees.

Three effects that we do not include act on the same scale. The electron loses up to 1.7 MeV by ionisation before it radiates ($dE/dx \simeq 0.35\,\text{MeV/cm}$ in LD_2), the CEBAF beam has an energy spread of order 10^{-4}, i.e. 1 MeV, and the bremsstrahlung on the atomic electron has its own endpoint $1.5\,m_e$ below E_0. Each smears or shifts the tip by about 1 MeV; none matters for the J/ψ yield. In the last 5 MeV the formula is also outside its domain, $E' \gg m_e$; that region holds less than 10^{-4} of the photons above 6.5 GeV.

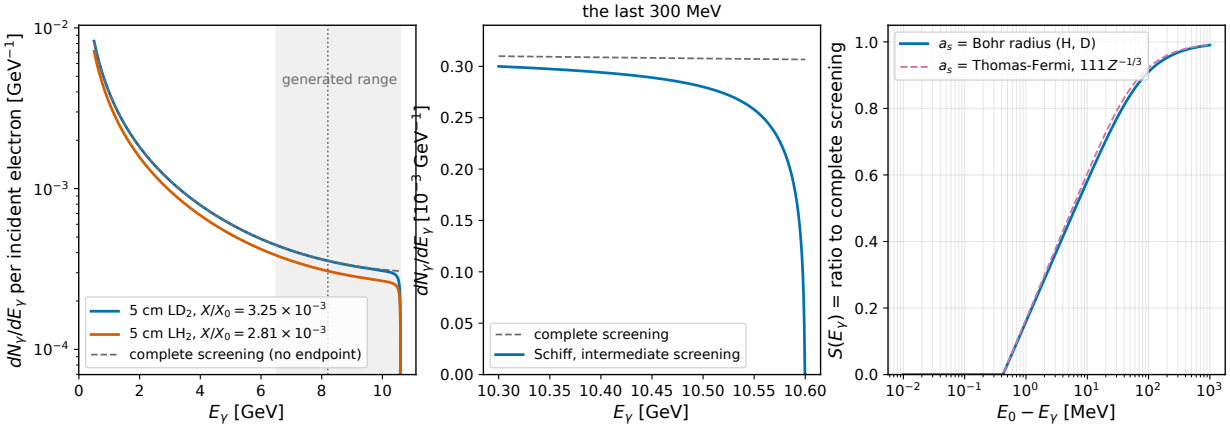


Figure 1: The photon beam. Left: photons per incident electron and per GeV from the 5 cm cell (average upstream thickness $L/2$), for LD_2 and LH_2; the dashed curve is the complete-screening formula without the endpoint factor; the shaded band is the generated range 6.5–10.6 GeV and the dotted line the free-neutron threshold. Middle: the last 300 MeV, where the Schiff intermediate screening spectrum bends down and reaches zero at the endpoint while the complete-screening formula stays finite. Right: the endpoint factor $S(E_\gamma)$ against the distance to the endpoint, for the Bohr radius used here and for Schiff's Thomas–Fermi radius.

3 The deuteron: spectator model

The deuteron is at rest in the laboratory. In the plane-wave spectator picture the photon interacts with one nucleon while the other is a spectator that leaves with the momentum it had in the nucleus. The spectator is on its mass shell, the struck nucleon carries the rest of the deuteron four-momentum and is off shell:

$$p_s = (E_s, \vec{p_s}), \quad E_s = \sqrt{M_p^2 + p_s^2}; \qquad p_N = (M_D - E_s, -\vec{p_s}), \qquad m^{*2} = p_N^2 < M_n^2. \tag{7}$$

The spectator momentum is drawn from the AV18 deuteron momentum distribution $n(p_s)$ (S and D waves, normalised to one, tabulated up to $p_s = 0.70\,\mathrm{GeV}/c$), isotropic in the laboratory. The photon–neutron invariant mass is built from the actual four-vectors,

$$s = (k + p_N)^2 = m^{*2} + 2E_\gamma\,(E_N - p_{N,z}), \tag{8}$$

so a neutron running toward the photon ($p_{N,z} < 0$) raises s and one running away lowers it. The free-neutron threshold $E_\gamma^{\mathrm{th}} = [(M_n + M_{J/\psi})^2 - M_n^2]/2M_n = 8.20\,\mathrm{GeV}$ is no longer a threshold: the reaction $\gamma d \to J/\psi\, n\, p$ is open from $[(M_{J/\psi} + M_n + M_p)^2 - M_D^2]/2M_D = 5.66\,\mathrm{GeV}$, and photons between 5.66 and 8.20 GeV produce J/ψ on the high-momentum tail of $n(p_s)$. The generated range starts at 6.5 GeV; below it the yield is negligible (Section 6.2).

Two conventions of the earlier electroproduction study are kept. The cross section is evaluated at the free-nucleon mass, as a function of the invariants s and t of the event, and m^* enters only where the event is built, i.e. in the kinematics of the final state. And a draw with $m^* < 0.10\,\mathrm{GeV}$ is rejected: there the struck nucleon becomes light-like and the plane-wave picture has no meaning. The floor was chosen for this study from the table of allowed configurations (Table 1): at 7.0 GeV, where only 0.9 % of the AV18 configurations can produce a J/ψ at all, the floor at 0.30 GeV used for π^0 would remove 3 % of them, the floor at 0.10 GeV removes 0.1 %.

Table 1: Kinematics of the struck neutron in the deuteron for a photon along $+z$ (AV18, 200 000 configurations, unweighted by the cross section): fraction of configurations above the J/ψ threshold, their mean spectator momentum, the fraction with $p_s > 300\,\mathrm{MeV}/c$, and the fraction of the allowed configurations removed by a floor on m^*.

E_γ [GeV]	allowed	$\langle p_s \rangle$ [MeV/c]	$p_s > 300$	floor 0.30	0.20	0.10
7.0	0.88 %	373	60.8 %	2.9 %	0.7 %	0.11 %
7.5	4.44 %	224	18.3 %	0.7 %	0.2 %	0.04 %
8.0	24.6 %	117	4.4 %	0.17 %	0.06 %	0.01 %
8.2	44.3 %	91	2.6 %	0.10 %	0.03 %	0.01 %
10.0	97.3 %	86	1.9 %	0.06 %	0.02 %	0.00 %

3.1 The event weight and the flux factor

Per incident electron and per target neutron, the number of J/ψ events is

$$dN = \frac{dN_\gamma}{dE_\gamma}\,dE_\gamma \;\times\; n(p_s)\,d^3p_s \;\times\; \frac{k \cdot p_N}{E_\gamma\,E_N}\,\sigma_{\gamma n}(s)\,, \qquad \frac{k \cdot p_N}{E_\gamma\,E_N} = 1 - \beta_{N,z}, \tag{9}$$

where the last factor is the Møller flux factor of the moving target: the rate of a massless projectile on a target with velocity $\vec{\beta}_N$ is σ times the relative velocity $1 - \beta_{N,z}$, not σ alone. For a nucleon at rest it is one, and averaged over the isotropic $n(p_s)$ it is also one. It is nevertheless not a detail here. A neutron running toward the photon has both a larger flux factor and a larger s, hence a larger cross section; the two are positively correlated and the average of the product exceeds the product of the averages. Because $\sigma_{\gamma n}(s)$ rises steeply above threshold (Section 4), this correlation is worth about +4 % on the deuteron yield at 10 GeV and more near threshold. It was absent from

the first version of the generator, and its absence produced a spurious 4 % deficit of the deuteron yield; the samples in this report include it, and Section 6.1 shows the ratio with and without it. The same factor is missing in the electroproduction branch of the generator (`generator/fermi.py` working in the struck-nucleon rest frame), where its numerical effect is much smaller because the π^0 and η cross sections vary slowly with W; it should be added there for consistency.

4 The elementary cross section

The hadronic side is the electroproduction model `jpsi.roberts2026` at $Q^2 = 0$, relabelled for a neutron target as `jpsin.roberts2026` with $\sigma_n = \sigma_p$. Nothing new is assumed on this side: in the Hand convention the transverse cross section at $Q^2 = 0$ *is* the real-photon cross section, so the generator uses the model's $\sigma_T(W, Q^2{=}0, t)$ as $d\sigma/dt$. The parametrisation is

$$\sigma_{\mathrm{tot}}(W) = A\,(W - W_{\mathrm{th}})^{\alpha}\, e^{-B\,(W - W_{\mathrm{th}})}, \qquad \frac{d\sigma}{dt} = \sigma_{\mathrm{tot}}(W)\,\frac{b\,e^{b\,t'}}{1 - e^{-b\,|t'|_{\max}}}, \quad t' = t - t_{\min}, \tag{10}$$

with $A = 1.018\,\mathrm{nb}$, $\alpha = 0.707$, $B = -0.928\,\mathrm{GeV}^{-1}$, $b = 1.67\,\mathrm{GeV}^{-2}$ and $W_{\mathrm{th}} = M_N + M_{J/\psi} = 4.036\,\mathrm{GeV}$. Between the free threshold and 10.6 GeV, $W - W_{\mathrm{th}}$ runs from 0 to 0.52 GeV and σ_{tot} from 0 to 1.05 nb (Figure 2); the logarithmic derivative $d\ln\sigma/dW = \alpha/(W - W_{\mathrm{th}}) - B$ is $2.7\,\mathrm{GeV}^{-1}$ at 10 GeV and diverges at threshold, which is what makes the deuteron yield sensitive to shifts of s of a few $10^{-2}\,\mathrm{GeV}^2$. The t range generated is $-6 \le t \le -0.001\,\mathrm{GeV}^2$, cut to the physical range at each W; $|t|_{\min}$ is 1.8, 0.62 and $0.36\,\mathrm{GeV}^2$ at 8.5, 9.5 and 10.5 GeV.

Figure 2 compares the parametrisation with the CLAS12 measurement on the proton [2] (Tables V and VI of its Appendix A). The total cross section is described within the errors from threshold to 9.6 GeV; above 9.7 GeV the data lie 20–30 % above the curve, one to two standard deviations per point. The t dependence of the data is compatible with the single exponential in t' at $|t| \lesssim 2\,\mathrm{GeV}^2$; the points beyond, at 3–4 GeV^2, sit above the exponential by a factor two to three. Neither deviation matters for the Fermi-motion ratio of Section 6.1, which depends on the slope of $\sigma(W)$ near threshold and on nothing else; it would matter for absolute yields, and a refit of A, α, B and b to these data is the natural next step.

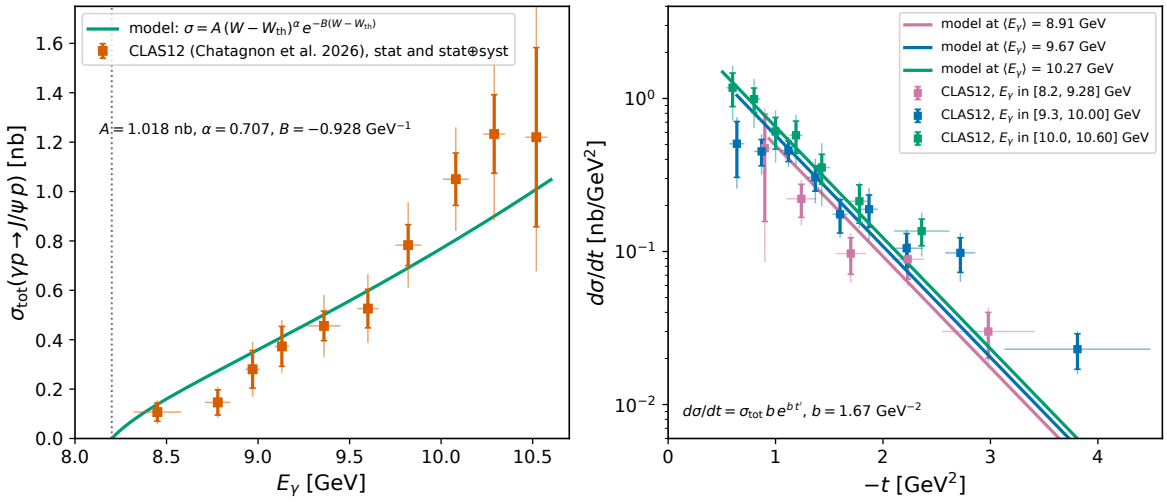


Figure 2: The elementary cross section used, compared with the CLAS12 measurement on the proton (Chatagnon et al. 2026, Appendix A tables). Left: $\sigma_{\mathrm{tot}}(\gamma p \to J/\psi\, p)$ of the `roberts2026` parametrisation against the photon energy on a nucleon at rest, with the CLAS12 total cross section in ten energy bins; the inner bar is the statistical error, the outer one statistical and systematic added in quadrature, the horizontal bar the rms of the photon energy in the bin. Right: $d\sigma/dt$ of the model at the mean photon energy of the three CLAS12 energy bins, with the CLAS12 differential cross sections of those bins (errors as on the left, horizontal bars the rms of $-t$ in the bin).

5 The generator and its validation

5.1 Algorithm

Each trial proceeds as follows.

1. E_γ is drawn from the spectrum (1) in $[k_{\min}, k_{\max}]$ by a tabulated inverse cumulative (20 001 points), including the endpoint factor.

2. The target nucleon is set at rest, or, on the deuteron, $\vec{p_s}$ is drawn from AV18, the struck four-vector and m^* are formed, and the trial is discarded if $m^* < 0.10\,\text{GeV}$ (0.01 % of the draws).

3. $s = (k + p_N)^2$ is computed; if it is below $(M_{J/\psi} + M_N)^2$ the trial is counted as below threshold and discarded.

4. The trial is accepted with probability $\sigma_{\text{tot}}(s)\,(1 - \beta_{N,z})/w_{\max}$. $\sigma_{\text{tot}}(s)$, the t integral of $d\sigma/dt$ over the card's range, is tabulated once on an 800-point grid in s. The majorant $w_{\max}$ is the maximum of $\sigma_{\text{tot}}(s)\,(1 - \beta_{N,z})$ along the head-on line, a neutron running toward the photon with p_s up to the AV18 ceiling at $k_{\max}$: for a given $|p_s|$ both s and the flux factor are largest there, so this is a strict bound and no event is under-weighted (the generator counts overweights; there were none). The price is efficiency: $w_{\max}/\langle\sigma\rangle \simeq 36$ on the deuteron and 4 on a free nucleon.

5. t is drawn from the exact inverse cumulative of $d\sigma/dt$ at the event's s (401 points), ϕ uniformly.

6. The J/ψ and the recoil neutron are built in the γN centre of mass and boosted to the laboratory with the actual photon and nucleon four-vectors, so on the deuteron $k + p_D = p_{J/\psi} + p_{n'} + p_s$ exactly. The J/ψ is decayed to e^+e^- isotropically. The vertex is drawn along the cell as in Section 2.

The events are written in LUND format with the photon as the beam (pid 22, energy E_γ in the header), the target pid 2112 or 1000010020, the J/ψ as a status-0 parent with its daughters, the recoil and, on the deuteron, the spectator proton. The run's `config.json` records the photon count per electron in the generated range, the flux-weighted $\langle\sigma\rangle$, and the yield per incident electron per target nucleon, $Y = \int dE_\gamma\,(dN_\gamma/dE_\gamma)\,\langle(1 - \beta_z)\sigma\rangle$.

5.2 Validation

- **Four-momentum conservation.** $|k + p_{\text{target}} - \sum p_{\text{final}}| < 5.5 \times 10^{-9}\,\text{GeV}$ for every event of the 200 000-event test samples, i.e. the ten significant digits of the LUND file; the spectator is on shell to 3×10^{-16}.

- **Photon spectrum.** On the free proton the generated E_γ distribution against the analytic $dN_\gamma/dE_\gamma \cdot \sigma_{\text{tot}}(W(E_\gamma))$: $\chi^2/\text{ndf} = 22.1/23$ over 6.5–10.6 GeV.

- **t slope.** An exponential fit of the generated t' distribution at $E_\gamma = 9.4$–$9.6\,\text{GeV}$ gives $b = 1.64\,\text{GeV}^{-2}$ against the model's 1.67 (the fit range includes the curvature of the physical limit).

- **Fermi kinematics.** The unweighted AV18 table of Table 1 is reproduced by the generator's own sampler; on the free neutron no event appears below 8.20 GeV, on the deuteron they do (Section 6.2).

- **Speed.** 0.09 ms per accepted event on a free nucleon and 0.65 ms on the deuteron on one core of the M4 Mac mini; the three 20-million-event samples took 35 minutes on ten cores, with no majorant violation in any of the 60 million events.

6 Results

6.1 Yield versus photon energy and the ratio deuteron / free neutron

Figure 3 shows the J/ψ yield per 100 MeV of photon energy, per incident electron and per neutron, on the deuteron and on a free neutron at rest, and their ratio above 8.5 GeV in 100 MeV bins. The per-electron normalisation is the same for both (same radiator, same dN_γ/dE_γ), so the ratio is the effect of Fermi motion and binding alone. With 20 million events per sample its statistical precision is 0.2 % per 100 MeV bin above 9 GeV.

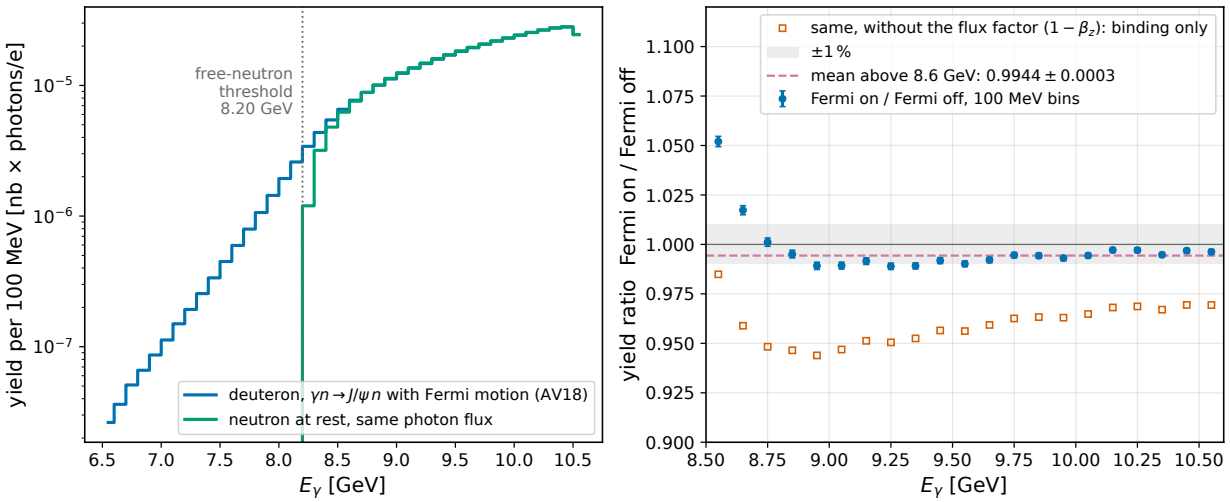


Figure 3: Left: J/ψ yield per 100 MeV of photon energy, per incident electron and per neutron, for the neutron in the deuteron (AV18) and for a free neutron at rest; the free threshold is at 8.20 GeV. Right: the ratio of the two above 8.5 GeV in 100 MeV bins (filled circles, statistical errors); the band is ±1 % and the dashed line the mean above 8.6 GeV. The open squares show the same ratio when the Møller flux factor $1-\beta_{N,z}$ of Eq. (9) is removed from the deuteron weights: this is the binding effect alone.

Table 2: Yield per incident electron and per neutron (10^{-6} nb×photons/e) in bins of photon energy, on the deuteron and on the free neutron, their ratio with the statistical error, and the ratio obtained without the flux factor $1-\beta_{N,z}$ (binding only).

E_γ [GeV]	deuteron	free neutron	ratio	ratio without flux factor
8.2–8.4	7.78	4.38	1.7757 ± 0.0047	1.6225
8.4–8.6	12.01	11.04	1.0879 ± 0.0020	1.0149
8.6–9.0	37.79	37.82	0.9992 ± 0.0010	0.9486
9.0–9.5	73.54	74.27	0.9902 ± 0.0007	0.9518
9.5–10.0	103.11	103.84	0.9929 ± 0.0006	0.9611
10.0–10.6	156.09	156.70	0.9961 ± 0.0005	0.9679
6.5–10.6 (all)	400.52	388.06	1.0321	0.9919

Table 2 gives the numbers. Above 8.6 GeV the ratio never leaves the band 0.99–1.00: it is 1.0173 ± 0.0023 in 8.6–8.7 GeV, passes through a shallow minimum of 0.9889 ± 0.0016 in 9.2–9.3 GeV, and comes back to 0.9961 in the last 600 MeV; its mean over 8.6–10.6 GeV is

$$\left.\frac{Y_d}{Y_n}\right|_{8.6<E_\gamma<10.6\,\mathrm{GeV}} = 0.9944 \pm 0.0003. \tag{11}$$

With 20 million events the statistical errors are small enough to resolve this sub-percent shape, which is why no straight line fits it; what matters is its size. The total yield per neutron over

the whole generated range is 1.0321 times the free-neutron one; the excess is the near-threshold region, 8.2–8.6 GeV, and the sub-threshold yield of Section 6.2. **Above 8.6 GeV Fermi motion does not change the J/ψ yield per neutron by more than 1 %.** This is the net of two effects that the last column of Table 2 separates. Without the flux factor the ratio is 0.9611 at 9.5–10 GeV: the struck neutron carries less than its rest energy ($\langle E_N \rangle = M_D - \langle E_s \rangle = 0.933$ GeV against $M_n = 0.940$ GeV) and is off shell ($\langle m^* \rangle = 0.915$ GeV), so at fixed E_γ the average s is lower than on a free neutron by about 0.16 GeV^2, i.e. W by 18 MeV, and with $d \ln \sigma / dW \simeq 2.7$ GeV^{-1} the yield drops by 4–5 %. The flux-factor correlation of Section 3.1 adds back +4 %. The two have different energy dependence, both growing toward threshold, which is why the ratio without the flux factor rises slowly with E_γ from 0.9486 to 0.9679 while the full ratio stays within 1 % of one.

Near the free threshold the picture changes completely. In 8.4–8.6 GeV the ratio is 1.0879, in 8.2–8.4 GeV it is 1.7757 (2.6 in the first 100 MeV above threshold), and below 8.20 GeV the free-neutron yield is zero while the deuteron yield is not: Fermi motion smears the sharp threshold of $\sigma_{\rm tot}(W) \propto (W - W_{\rm th})^{0.7}$ over ±300 MeV of photon energy. Any extraction of the near-threshold cross section from deuterium data has to unfold this, and the generator provides the response.

6.2 The yield below the free threshold

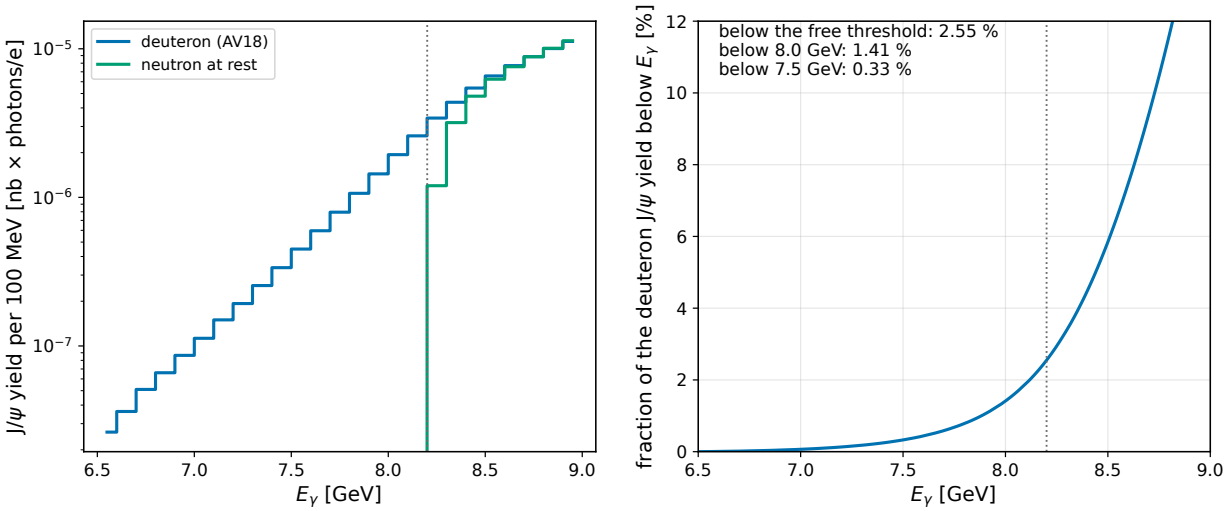


Figure 4: Left: the yield per 100 MeV between 6.5 and 9 GeV on the deuteron and on the free neutron (logarithmic scale). Right: cumulative fraction of the deuteron J/ψ yield below a given photon energy.

Figure 4 shows the region below and around the free threshold. On the deuteron 2.55 % of the J/ψ yield in the generated range, 10.21×10^{-6} nb× photons per electron, is produced by photons below 8.20 GeV; 1.41 % below 8.0 GeV and 0.33 % below 7.5 GeV. The sample contains 13282, 52273, 216927 and 227595 events in 6.5–7.0, 7.0–7.5, 7.5–8.0 and 8.0–8.2 GeV. The yield falls by an order of magnitude per 500 MeV below threshold, following the high-momentum tail of $n(p_s)$ folded with the threshold rise of the cross section, and it depends on exactly the ingredients that are least certain: the AV18 tail beyond 300 MeV/c, the off-shell prescription and the absence of final-state interactions. The numbers below 8 GeV should be read as the plane-wave expectation, good to a factor of order two rather than to percent.

6.3 Spectator momenta and the off-shell mass

The spectator proton is written in every deuteron event and Figure 5 shows what it looks like. Above the free threshold the spectators follow the bulk of AV18: $\langle p_s \rangle = 99$ MeV/c and 4.4 % above 300 MeV/c, with $\langle m^* \rangle = 0.915$ GeV. Below threshold the events are selected from the tail: $\langle p_s \rangle = 229$ MeV/c and 26.2 % above 300 MeV/c for all sub-threshold events, 278 MeV/c and 36.2 % below 8.0 GeV, 400 MeV/c and 67.6 % below 7.5 GeV, with $\langle m^* \rangle$ falling to 0.706 GeV. The floor

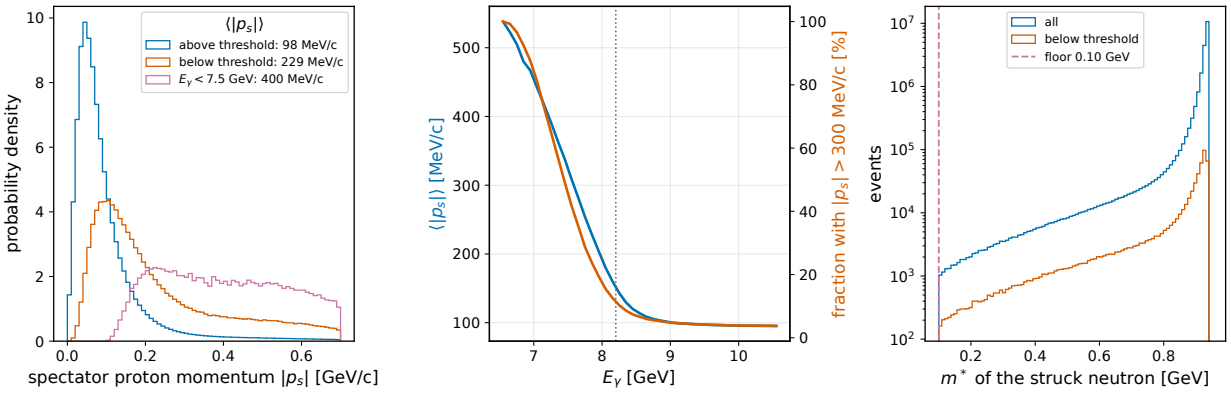


Figure 5: Left: distribution of the spectator proton momentum for events above the free threshold, below it, and below 7.5 GeV. Middle: mean spectator momentum and the fraction of events with $p_s > 300\,\mathrm{MeV}/c$ against the photon energy. Right: the off-shell mass m^* of the struck neutron, all events and sub-threshold events; the floor at 0.10 GeV is indicated.

at 0.10 GeV removes 0.01 % of the draws overall and is invisible in any of these distributions. A sub-threshold J/ψ comes with a fast spectator proton at large angle that CLAS12 can detect, which is the experimental handle on this part of the sample.

6.4 The t distributions

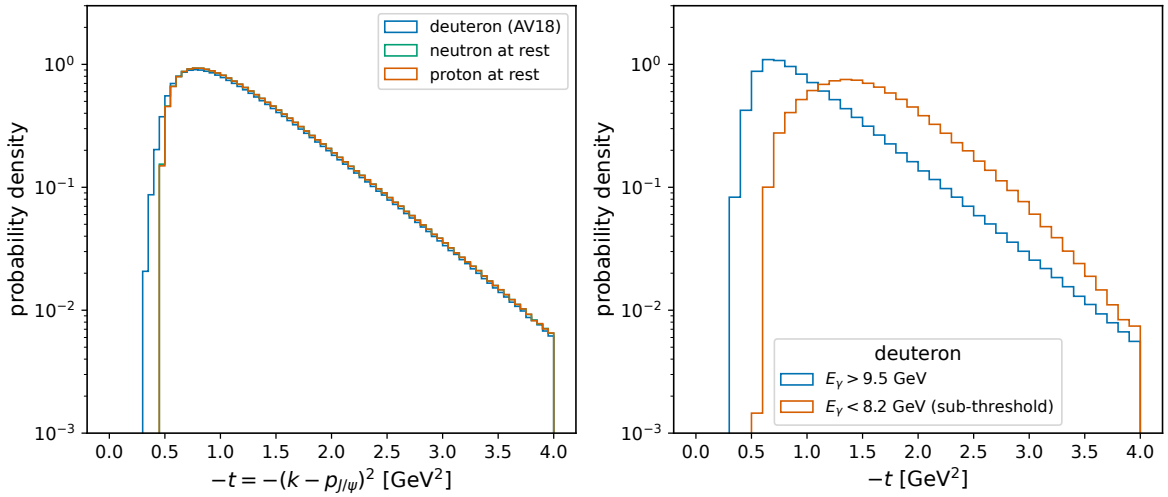


Figure 6: Left: $-t = -(k - p_{J/\psi})^2$, with k the photon four-vector, for the three samples. Right: on the deuteron, above 9.5 GeV and below the free threshold.

Figure 6 shows $-t = -(k - p_{J/\psi})^2$, computed as an analysis would from the photon and the J/ψ, without knowledge of the spectator. Above threshold the deuteron and the free-nucleon distributions coincide: the t dependence is generated at the event's own s and Fermi motion only redistributes events among nearby s. Below the free threshold the events sit at the edge of phase space, $|t|_{\min}$ is large and the distribution is compressed toward it; the t slope of a sub-threshold sample is not the elementary slope b.

7 Which variable for the cross section: W, not E_γ

On a free nucleon the photon energy and the invariant mass are the same variable, $W^2 = M^2 + 2ME_\gamma$, and the free-proton measurements are quoted against E_γ by convention. On the deuteron the map

is gone: the elementary cross section is a function of $s = (k + p_N)^2$ with a moving, bound nucleon, and one lab photon energy feeds a band of W. This section establishes, with the generator, which quantities an analysis of $\gamma d \to J/\psi\, n\, (p_s)$ can reconstruct and which one the cross section should be presented against.

7.1 The photon energy from the final state

With the J/ψ and the neutron measured and the deuteron at rest, the undetected spectator proton closes the kinematics. Requiring it to be on its mass shell, $(k + P_d - p_\psi - p_n)^2 = M_p^2$ with $k = (E_\gamma, 0, 0, E_\gamma)$, gives

$$E_\gamma = \frac{(E_\psi + E_n - M_d)^2 - |\vec{p}_\psi + \vec{p}_n|^2 - M_p^2}{2\,(E_\psi + E_n - p_{z,\psi} - p_{z,n} - M_d)}\,, \tag{12}$$

Eq. (10) of Ref. [1] with proton and neutron exchanged. It is exact for a real photon along the beam, independent of the Fermi-motion model, and it also yields the spectator momentum $\vec{p}_s = \vec{k} - \vec{p}_\psi - \vec{p}_n$ and hence the off-shell mass $m^{*2} = (M_d - E_s)^2 - p_s^2$ of the struck neutron, event by event. For a quasi-real virtual photon the same derivation gives a correction $+Q^2/2(E_s - p_{z,s}) \simeq 0.53\,Q^2$, i.e. 5 MeV at $Q^2 = 0.01\,\mathrm{GeV}^2$ and 50 MeV at $0.1\,\mathrm{GeV}^2$.

Figure 7 compares Eq. (12) with four other ways of obtaining a photon energy from the final state, on the deuteron sample where the true E_γ is known (Table 3). Energy conservation with a neutron at rest, $E_\psi + E_n - M_n$, is low by the spectator kinetic energy plus the binding energy, 12 MeV with a 24 MeV spread; longitudinal momentum conservation, $p_{z,\psi} + p_{z,n}$, is off by the spectator's p_z; the J/ψ alone, as in an LH_2 analysis, carries the full Fermi smearing of 200 MeV. The last entry, $(s - M_n^2)/2M_n$ with $s = (p_\psi + p_n)^2$, is not an estimator of the lab photon energy at all: it is the photon energy a free neutron would need to reach the measured W, and it differs from the true E_γ by the Fermi smearing of s itself, 0.67 GeV rms. With detector resolution (illustrative: 1 % and 1 mrad for the J/ψ, 10 % and 10 mrad for the neutron) Eq. (12) and the energy-conservation estimate are both limited by the neutron momentum, 135 MeV rms, which is the mechanism behind the 41–86 MeV resolution Ref. [1] finds for SoLID in the proton channel.

Table 3: Mean shift and rms of five photon-energy prescriptions against the true E_γ, deuteron events above 8.6 GeV, ideal and smeared momenta (MeV).

	ideal		smeared	
prescription	shift	rms	shift	rms
Eq. (12), spectator on shell	0	0	+3	133
$E_\psi + E_n - M_n$	−12	24	−10	136
$p_{z,\psi} + p_{z,n}$	−38	85	−38	171
J/ψ alone, neutron at rest	+40	196	+41	209
$(s - M_n^2)/2M_n$ (not an estimator of E_γ)	+245	667	+262	684

7.2 W rules the cross section

Figures 8 and 9 show the final-state invariant mass against the lab photon energy of Eq. (12), once as W and once converted to the equivalent free-neutron photon energy $E_\gamma^{\mathrm{eff}} = (W^2 - M_n^2)/2M_n$. On a free neutron every event would lie on the orange line. On the deuteron one photon energy feeds a band 0.13 GeV wide in W (0.6–0.7 GeV in E_γ^{eff}); the band's mean runs 40–50 MeV in W above the free line because the flux-factor selection favours nucleons moving toward the photon; and below the free threshold the band does not start at the line at all. The photons of 6.7–8.2 GeV produce the reaction at $W = 4.17$–4.22 GeV, about 150 MeV above $M_n + M_{J/\psi}$, the same W a 9 GeV photon gives on a free neutron. The elementary process is ordinary there; only the photon is

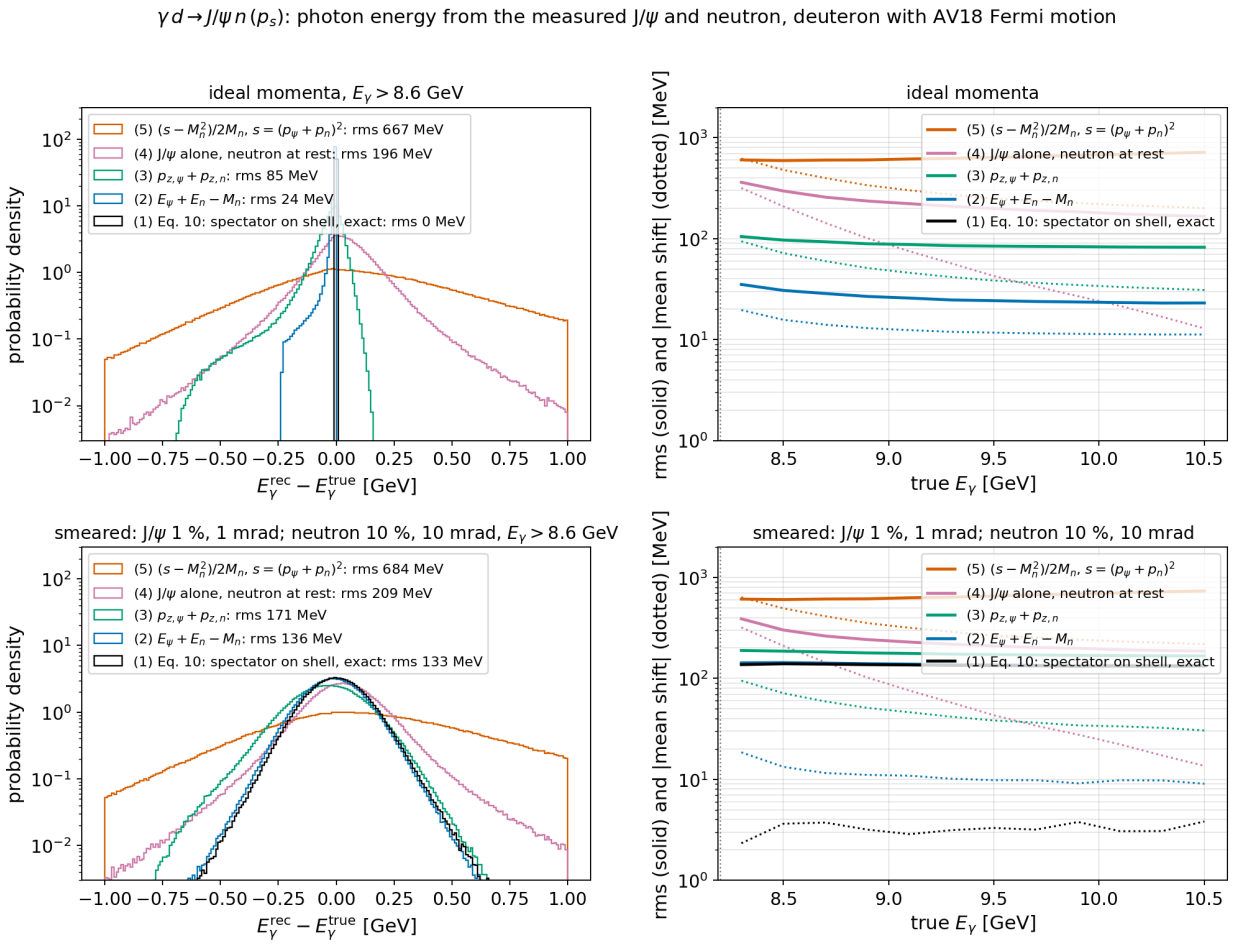


Figure 7: Photon energy reconstructed from the measured J/ψ and neutron on the deuteron, minus the true photon energy of the generator, for five prescriptions; left: distributions above 8.6 GeV, right: rms (solid) and mean shift (dotted) against the true E_γ. Top: ideal momenta; bottom: with an illustrative detector resolution.

sub-threshold. At the other end, photons of 10.5 GeV reach W values a free neutron would need 10.7 ± 0.7 GeV for, and 19 % of all deuteron events lie beyond the free-nucleon endpoint. A cut $|p_s| < 0.15\,\mathrm{GeV}/c$ narrows the band by a third and brings its mean to within 20 MeV of the free line (right panels).

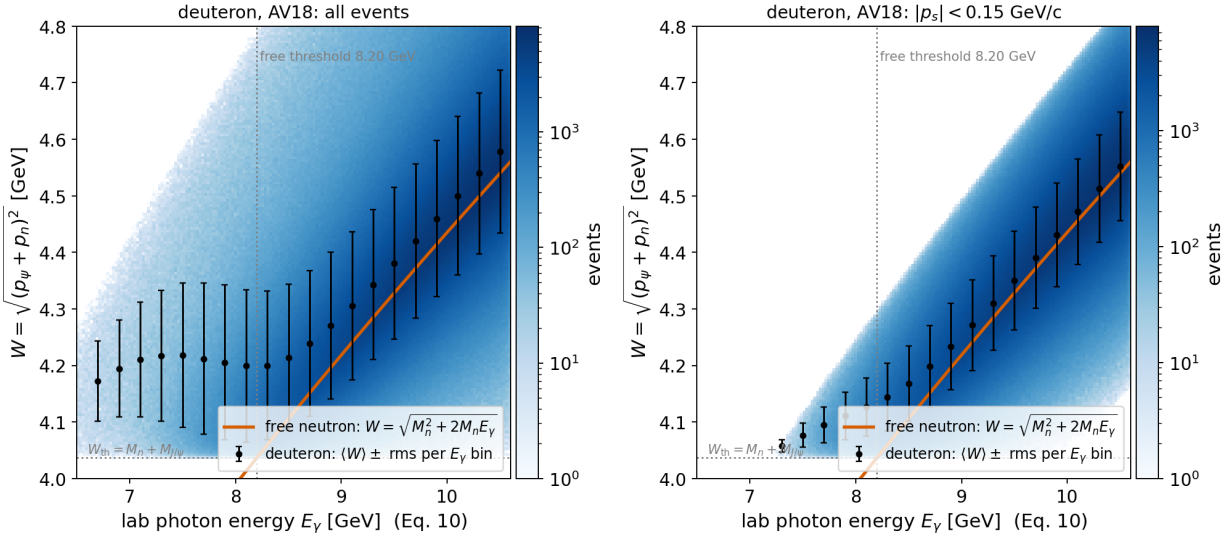


Figure 8: W from the measured J/ψ and neutron against the lab photon energy of Eq. (12), deuteron with AV18; orange: the free-neutron relation; black: mean and rms of W per 200 MeV bin of E_γ. Left: all events; right: $|p_s| < 0.15\,\mathrm{GeV}/c$.

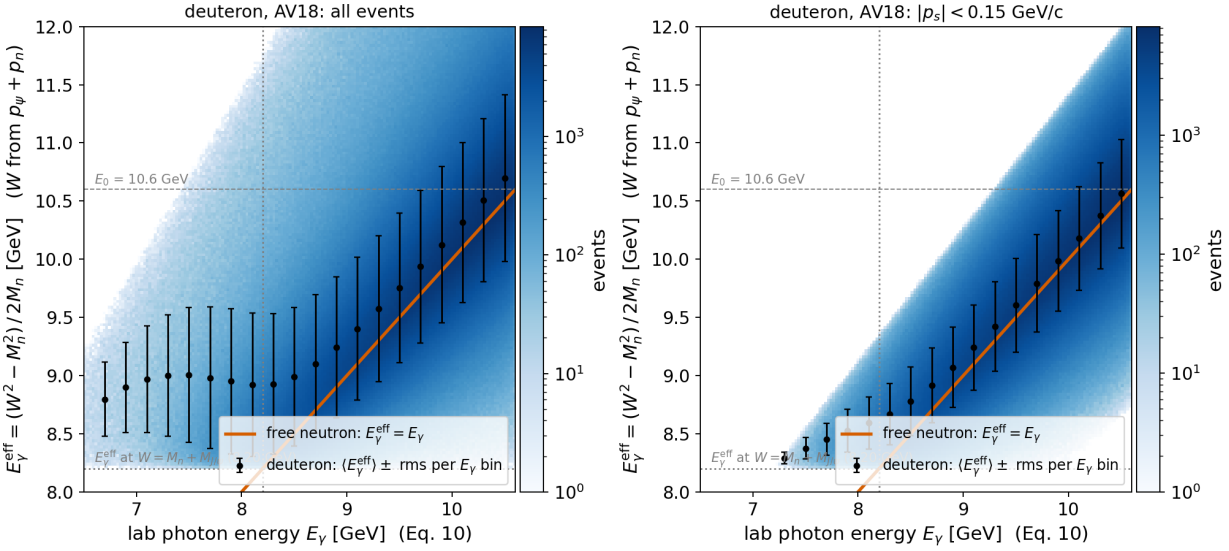


Figure 9: The same in photon-energy units: $E_\gamma^{\mathrm{eff}} = (W^2 - M_n^2)/2M_n$ against the lab E_γ.

The cross section therefore has to be presented against W, the variable it is a function of and the one the final state measures exactly, with no assumption about the target. The lab photon energy becomes a label: for each W bin one quotes $\langle E_\gamma \rangle$ and its spread (Table 4), as Ref. [2] does for its bins. That spread is physics, 0.5 GeV rms, not resolution. Above 10 GeV the mean freezes at 10.15 GeV because $E_\gamma \leq E_0$: those W bins are reached only by the upper half of the Fermi distribution.

Table 4: Lab photon energy per bin of W on the deuteron: mean, rms and 16–84 % range of E_γ from Eq. (12), all events and with $|p_s| < 0.15\,\mathrm{GeV}/c$.

W [GeV]	E_γ^{eff} [GeV]	$\langle E_\gamma\rangle\pm$rms, all	16–84 %	$\langle E_\gamma\rangle\pm$rms, $\lvert p_s\rvert < 0.15$
4.036–4.083	8.2–8.4	8.39 ± 0.51	7.97–8.81	8.39 ± 0.40
4.128–4.174	8.6–8.8	8.75 ± 0.52	8.32–9.19	8.77 ± 0.42
4.218–4.263	9.0–9.2	9.12 ± 0.51	8.69–9.58	9.16 ± 0.42
4.307–4.350	9.4–9.6	9.48 ± 0.49	9.07–9.93	9.53 ± 0.40
4.393–4.436	9.8–10.0	9.81 ± 0.45	9.42–10.23	9.87 ± 0.36
4.478–4.519	10.2–10.4	10.06 ± 0.42	9.71–10.43	10.13 ± 0.30
4.561–4.602	10.6–10.8	10.17 ± 0.41	9.84–10.50	10.28 ± 0.25
4.643–4.683	11.0–11.2	10.16 ± 0.44	9.81–10.51	10.35 ± 0.19

7.3 The effective photon flux in W

The price of binning in W is the flux: bremsstrahlung photons are counted per lab E_γ, and Fermi motion redistributes them in W. The number of photons that produced a given W,

$$\Phi_d(W)\,dW = \int dE_\gamma\,\frac{dN_\gamma}{dE_\gamma}\int d^3p_s\,n(p_s)\,(1-\beta_z)\,\delta\big(W - W(E_\gamma,\vec{p}_s)\big)\,dW, \tag{13}$$

is obtained from the generator by weighting every event with $1/\sigma(s)$. Figure 10 compares it with the free-neutron flux in the same W bin (Table 5). Up to 9 GeV the two differ by 2–3 %, and the difference is insensitive to the Fermi model because the core of the wave function determines it; toward the endpoint it grows to 10–40 %, and beyond $E_\gamma^{\mathrm{eff}} = E_0$ the free flux is zero while the deuteron flux is not. The spectator cut lowers Φ_d by the 5–6 % of events it removes and does not change the shape. The cross section extracted from a W bin is the acceptance-corrected yield divided by $\Phi_d(W)$ and by the number of neutrons; the generator provides Φ_d for any selection, and its model dependence (wave function, off-shell prescription) is the systematic of the method.

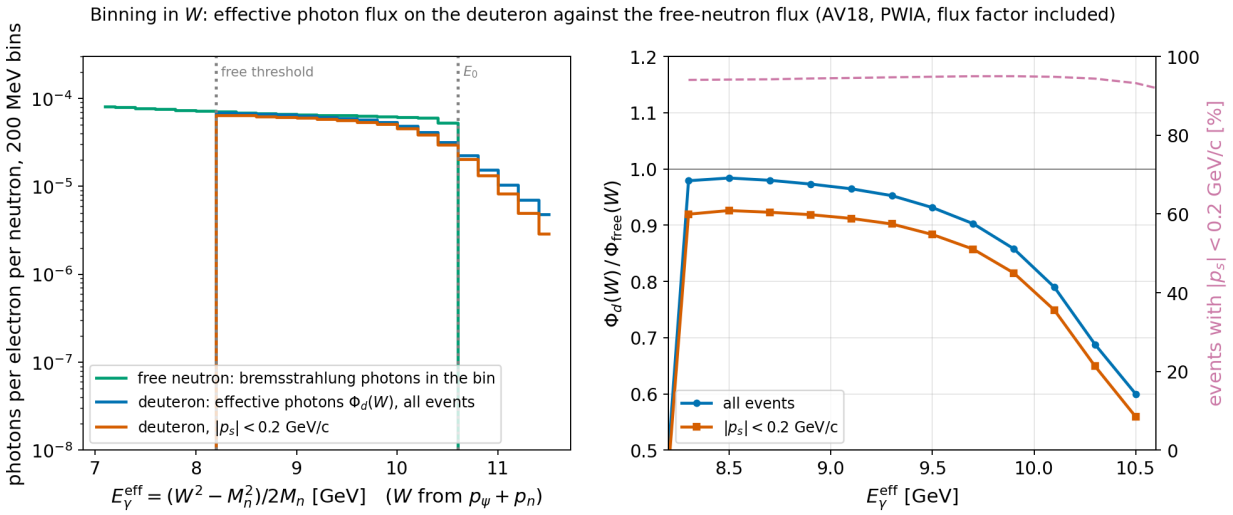


Figure 10: Left: photons per electron and per neutron in bins of E_γ^{eff}, i.e. of W: the bremsstrahlung flux on a free neutron and the effective flux $\Phi_d(W)$ on the deuteron, all events and with $|p_s| < 0.15\,\mathrm{GeV}/c$. Right: their ratio, and the fraction of events inside the spectator cut.

7.4 The struck nucleon is off shell

What is measured is $\gamma\,n^* \to J/\psi\,n$ with $m^* < M_n$, and a cross section for an off-shell nucleon is not an observable: the connection to the free reaction is a prescription. The deuteron keeps

Table 5: Effective flux on the deuteron relative to the free-neutron flux in the same W bin.

E_γ^{eff} [GeV]	8.3	8.9	9.3	9.7	10.1	10.5	> 10.6
$\Phi_d/\Phi_{\mathrm{free}}$, all	0.98	0.97	0.95	0.90	0.79	0.60	∞
$\Phi_d/\Phi_{\mathrm{free}}$, $\lvert p_s\rvert < 0.15$	0.92	0.92	0.90	0.86	0.75	0.56	∞

the prescription under control because the off-shellness is small and measured (Table 6). Within $|p_s| < 0.15\,\mathrm{GeV}/c$, 84 % of the events, the struck neutron is off shell by 8 MeV on average and 26 MeV at most, and the ambiguity of where to evaluate $\sigma(s)$ is $(M_n^2 - m^{*2})/(s - M_n^2) \simeq 10^{-3}$, equivalent to 5–10 MeV on the E_γ^{eff} axis. The generator's prescription, σ_{free} at the measured invariants with m^* in the kinematics only, and its alternatives (the flux or Hand factor with m^*, the on-shell energy for the struck nucleon) differ by that amount in the bulk and by 2–3 % at $p_s = 0.3\,\mathrm{GeV}/c$. The formally correct limit is the deuteron pole, $m^{*2} \to M_n^2$, where the residue is the free cross section and final-state interactions vanish; measuring the extracted σ in two or three bins of p_s and checking that it does not move is the practical form of that extrapolation. Final-state interactions are outside the generator: J/ψ–nucleon rescattering is below 1 % except near threshold, where it is the physics of interest, and nucleon–nucleon rescattering is confined to $p_s > 0.3\,\mathrm{GeV}/c$.

Table 6: Off-shellness of the struck neutron as a function of a spectator-momentum cut, events above the free threshold.

$\lvert p_s\rvert$ cut [GeV/c]	kept	$\langle M_n - m^*\rangle$ [MeV]	max $M_n - m^*$ [MeV]	$(M_n^2 - m^{*2})/(s - M_n^2)$
< 0.05	32 %	3.5	5	4×10^{-4}
< 0.10	68 %	6	13	6×10^{-4}
< 0.15	84 %	8	26	8×10^{-4}
< 0.20	91 %	10	45	1.0×10^{-3}
all	100 %	24	840	2×10^{-3}

7.5 Recommendation

Reconstruct E_γ and $\vec{p}_s$ event by event with Eq. (12). Select $|p_s| < 0.15\,\mathrm{GeV}/c$. Bin in W from $p_\psi + p_n$. Divide the acceptance-corrected yield by $\Phi_d(W)$ from the generator with the same selection. Present the result as $\sigma(\gamma n \to J/\psi\, n)$ against W, with $\langle E_\gamma\rangle$ and its spread per bin as a second column, together with the free-proton data at the same W. Quote as systematics the prescription spread, the wave-function dependence of Φ_d inside the cut, and the stability of σ against p_s. Show the full sample separately as the sub-threshold and large-p_s physics, without calling it the free cross section. Do not use $\langle E_\gamma\rangle$ as the abscissa of the cross section, and do not call $(s - M_n^2)/2M_n$ a photon energy.

8 Discussion and limits

What the result means. For the J/ψ yield per neutron above 8.6 GeV, a generator without Fermi motion is as good as one with it, to better than 1 % at every photon energy. This is not because Fermi motion is small (the smearing of s is $\pm 0.7\,\mathrm{GeV}^2$) but because its two first-order effects on a steeply rising cross section, binding and the flux-factor correlation, cancel to that level for the AV18 deuteron. The cancellation is specific to the observable (yield at fixed lab photon energy, integrated over the spectator) and to the steepness of $\sigma(W)$; it would not hold for a different α or for a target with larger binding.

Where Fermi motion is essential. Within 400 MeV of the free threshold the ratio departs from one by tens of percent and below 8.20 GeV the whole yield is a Fermi-motion effect. The E_γ-dependence of a threshold measurement on deuterium must be unfolded with this response, and the spectator proton, when detected, tags the events that carry it.

8.1 Do we need a Monte Carlo with Fermi motion?

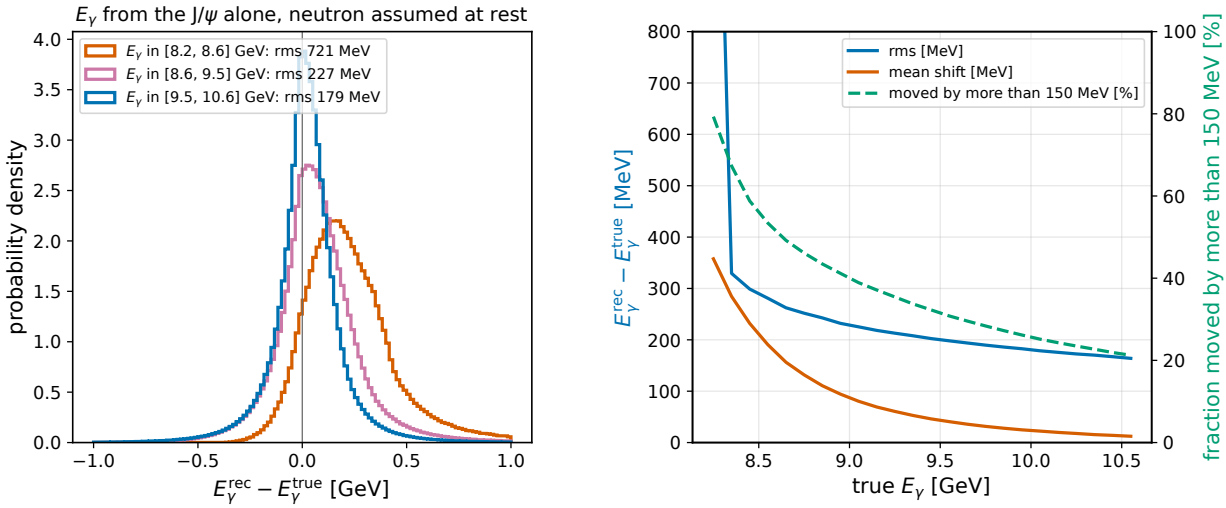


Figure 11: Photon energy reconstructed from the J/ψ four-vector alone with the target neutron assumed at rest, as an untagged analysis does it, minus the true photon energy, on the deuteron. Left: distributions in three ranges of true E_γ. Right: rms and mean shift, and the fraction of events displaced by more than 150 MeV, against the true E_γ. On a free neutron all three are zero.

The answer has two parts, and they point in different directions.

For the yield per neutron at the true photon energy, no. Above 8.6 GeV a free-neutron generator reproduces the deuteron yield to better than 1 % in every 100 MeV bin (Table 2), and the t distribution at the true s is the same. If the analysis knew the photon energy and the momentum transfer event by event, Fermi motion could be switched off.

For everything an analysis actually measures on deuterium, yes. An untagged analysis does not know E_γ; it reconstructs it from the final state with the target at rest, and Fermi motion enters that reconstruction at full strength. Figure 11 and Table 7 show the photon energy obtained from the J/ψ alone, $E_\gamma^{\rm rec} = (2M_n E_{J/\psi} - M_{J/\psi}^2)/2(M_n - E_{J/\psi} + p_{z,J/\psi})$: it is smeared by 196 MeV rms above 8.6 GeV, shifted upward by +120 MeV in 8.6–9.0 GeV and by +252 MeV in the bins next to threshold, and 29 % of the events move by more than 150 MeV, i.e. out of a typical energy bin. The momentum transfer computed with the reconstructed photon is smeared by 0.24 GeV^2, comparable to $1/b$. A free-neutron Monte Carlo has none of this: it would give the wrong bin migration, the wrong t resolution, the wrong efficiency of any exclusivity or coplanarity cut, and no events below the free threshold or with a fast spectator. The near-threshold region, where the ratio is 1.09 and 1.78 and where the physics interest lies, is entirely a Fermi-motion effect.

Since the deuteron generator costs 0.65 ms per event and passes every acceptance test, there is no reason to generate deuterium without it. The free-neutron sample keeps one role: the cross-check that isolates the Fermi-motion effect, as in this report.

Limits of the model. The calculation is plane-wave impulse approximation with a spectator on shell and the cross section evaluated at the free-nucleon mass. Not included: final-state interactions of the J/ψ or the recoil neutron with the spectator, coherent production on the deuteron, any modification of the elementary amplitude off shell, the D-wave tensor structure (only $|\psi|^2$ enters), and the difference between σ_n and σ_p. The AV18 tail above 300 MeV/c is used as

Table 7: Resolution of the photon energy reconstructed from the J/ψ alone on the deuteron (neutron assumed at rest): mean shift, rms, fraction of events moved by more than 150 MeV, and the rms of the momentum transfer computed with the reconstructed photon minus the true one.

true E_γ [GeV]	shift [MeV]	rms [MeV]	moved > 150 MeV	Δt rms [GeV^2]
8.2–8.6	+252	721	62 %	1.31
8.6–9.0	+120	247	45 %	0.34
9.0–9.5	+60	213	35 %	0.27
9.5–10.0	+32	190	28 %	0.23
10.0–10.6	+17	171	23 %	0.19
8.6–10.6	+40	196	29 %	0.24

is. The bremsstrahlung is thin-target and does not include the energy loss before radiation or the beam energy spread, each of order 1 MeV at the endpoint. The elementary cross section is one parametrisation, `roberts2026`, with $b = 1.67\,\mathrm{GeV}^{-2}$ independent of W; the ratio of Section 6.1 depends on its slope near threshold, not on its normalisation.

References

[1] T. Liu, Z. Zhao, M. Cai, D. Byer, H. Gao, *Subthreshold production of J/ψ mesons from the deuteron with the proposed Solenoidal Large Intensity Device*, Phys. Rev. C **109**, 065206 (2024).

[2] P. Chatagnon et al. (CLAS Collaboration), *Measurement of the near-threshold J/ψ photoproduction cross section with the CLAS12 experiment*, arXiv:2602.22128 [hep-ex] (2026), JLAB-PHY-26-4605, submitted to Phys. Rev. C.

A Cards, commands and files

The deuteron card (`photo20M/card_d2.txt`); the neutron and proton cards differ only in the model (`jpsin.roberts2026` with `fermi = none`; `jpsi.roberts2026`), in `target_X0_cm = 890.4` for LH_2 and in the output directory:

```
model = jpsin.roberts2026
ebeam = 10.6
kmin  = 6.5
kmax  = 10.6
tmin  = -6.0
tmax  = -0.001
nev   = 20000000
seed  = 2027
fermi = av18
fermi_mstar_min = 0.10
target_length_cm = 5.0
target_X0_cm = 769.1
vz = -5.5 -0.5
idecay = 1
maj_safety = 1.05
outroot = ROOT/photo20M/d2
```

Generation, figures and this document:

```
python3 -m exclurad_py.generator.photo card_d2.txt --jobs 10
```

```
python3 validation/jpsi_photo_report_figs.py ROOT/photo20M ROOT/photo20M/report
python3 validation/numbers_to_tex.py ROOT/photo20M/report/numbers.json numbers.tex
pdflatex jpsi_fermi_photoproduction.tex
```

The samples (LUND files, 42 GB) are on the Mac mini, with `ROOT` the directory `/Volumes/wd_14tb/aao_norad_kin/jpsi_fermi`; each run directory carries its `inputs/config.json`. The figures, `numbers.json` and this report are in the `jpsi-fermi` branch under `docs/jpsi_fermi/report/`, and in the OneDrive folder `Work/2026_Jpsi_Fermi_photoproduction`.